

Spores *to* seeds

Although they are rooted in the ground, plants find ways of spreading their offspring. Mosses and ferns do it by scattering dustlike spores. However, most plants do it with seeds. Inside each seed is a tiny baby plant that even comes with its own store of food. This gives it a better chance of surviving and germinating (developing) when it reaches moist ground.

Mosses and ferns

The first plants to evolve on land reproduced using spores. Each dustlike spore is a tiny single cell that can germinate into a new plant. Mosses produce spores in capsules on stalks, while most ferns make spores on the underside of their leaves.

Ferns grow clusters of tiny brown cases. As the cases dry out and split open, the spores inside burst into the air.



Moss spores are released from tall stalks so they are more likely to drift away on the wind.



Most fern leaves unfurl from buds called fiddleheads.



Growing from seeds

Most plants produce dustlike pollen. However, unlike spores, pollen does not develop on its own but is used to fertilize the plant's eggs. Each fertilized egg then grows into a tiny plant embryo inside a seed.



Some plants, such as pine trees, produce pollen and eggs inside cones.



Most cones turn hard and woody as they mature, but the cones of junipers develop into sweet berries.



Many cone-producing plants have tough needlelike leaves, but those of the monkey puzzle tree are wide and fleshy.